$\boxed{T5}$ ① $\xi \sim R(\theta, 2\theta)$

$g(x,\theta) = \frac{1}{\theta}\{(\theta, 2\theta)\}$

$\theta > 0$

$M\xi = \int_\theta^{2\theta} \frac{x\,dx}{\theta} = \boxed{\frac{3}{2}\theta}$

$M\xi^2 = \int_\theta^{2\theta} \frac{x^2}{\theta}dx = \boxed{\frac{7\theta^2}{3}}$; $D\xi = M\xi^2 - (M\xi)^2 = \frac{7\theta^2}{3} - \frac{9\theta^2}{4} = \boxed{\frac{\theta^2}{12}}$

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а)

$\alpha_1 = \frac{3}{2}\theta$; $\mu_2 = \frac{\theta^2}{12}$; ОММ: $\tilde{\alpha}_1 = \alpha_1(\theta)$; $\frac{3}{2}\theta = \bar{x} \Rightarrow \tilde{\theta}_1 = \frac{2}{3}\bar{x}$ ~ ОММ

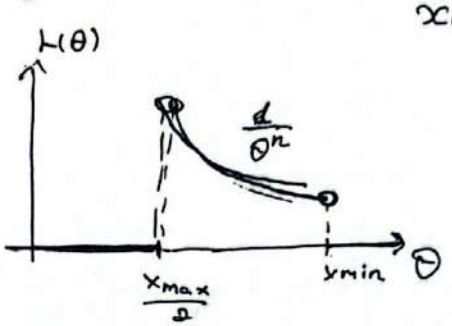
$M\tilde{\theta}_1 = \theta \Rightarrow$ несмещенная

$D\tilde{\theta}_1 = \frac{4}{9n}D\xi = \frac{4\theta^2}{108n} = \frac{\theta^2}{27n} \xrightarrow[n\to\infty]{} 0 \Rightarrow \tilde{\theta}_1$ - состоятельная

ОМП: $L(\theta) = \prod_{i=1}^{n} g(x,\theta) \Rightarrow L(\theta) = \frac{1}{\theta^n}\{(0 < x_i < 2\theta)\} \Rightarrow$ $x_{max} < 2\theta$, $x_{min} > \theta$ $\Rightarrow$

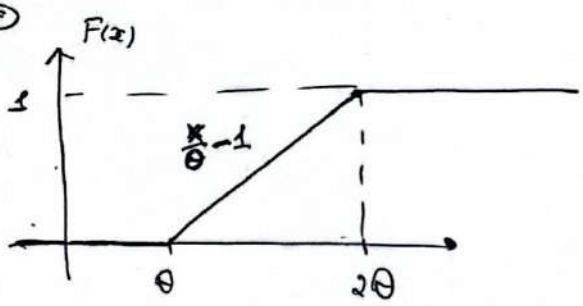
$\Rightarrow \theta \in \left(\frac{x_{max}}{2}, x_{min}\right)$

$\Rightarrow \tilde{\theta}_2 = \frac{x_{max}}{2}$ - ОМП



$M\tilde{\theta}_2 = M\frac{x_{max}}{2} = \frac{1}{2}\int_\theta^{2\theta} xn\left(\frac{x}{\theta}-1\right)^{n-1}\frac{1}{\theta}dx =$

$x_{max} \sim (F(x))^n = \psi(x)$



$\psi(x) = \psi'(x) = n\left(\frac{x}{\theta}-1\right)^{n-1}\cdot\frac{1}{\theta}$

$= \frac{n\theta}{2}\int_0^1 (1+y)y^{n-1}dy = \frac{n\theta}{2}\int_0^1 (y^{n-1}+y^n)dy = \frac{n\theta}{2}\left(\frac{1}{n}+\frac{1}{n+1}\right) = \frac{\theta(2n+1)}{2(n+1)} \Rightarrow \tilde{\theta}_2$ - смещенная

$y = \left(\frac{x}{\theta}-1\right)$

$\tilde{\theta}_2' = \frac{2(n+1)}{2n+1}\tilde{\theta}_2 = \frac{n+1}{2n+1}x_{max} \Rightarrow \tilde{\theta}_2'$ ~ несмещенная

Состоятельность $\tilde{\theta}_2$:

$M\left[\frac{x_{max}}{2}\right]^2 = \frac{n\theta^2}{4}\int_0^1 (1+y)^2 y^{n-1}dy = \frac{n\theta^2}{4}\left(\frac{1}{n}+\frac{2}{n+1}+\frac{1}{n+2}\right) =$

$= \frac{\theta^2}{4}\frac{(n+1)(n+2)+2n(n+2)+n(n+1)}{(n+1)(n+2)} = \frac{\theta^2(4n^2+8n+2)}{4(n+1)(n+2)}$

$D\tilde{\theta}_2 = M\tilde{\theta}_2^2 - (M\tilde{\theta}_2)^2 = \frac{\theta^2(4n^2+8n+2)}{4(n+1)(n+2)} - \frac{\theta^2(4n^2+4n+1)}{4(n+1)^2} = \frac{\theta^2}{4}\left(\frac{(4n^2+8n+2)(n+1) - (4n^2+4n+1)(n+2)}{(n+1)^2(n+2)}\right) = \frac{n\theta^2}{4(n+1)^2(n+2)} \xrightarrow[n\to\infty]{} 0 \Rightarrow \tilde{\theta}_2$ - состоятельная

②

Состоятельность $\widetilde{\Theta}_2'$; $D\widetilde{\Theta}_2' = \underbrace{\frac{(2n+2)^2}{(2n+1)^2}}_{\to 1} \underbrace{D\widetilde{\Theta}_2}_{\to 0} \xrightarrow[n\to\infty]{} 0 \Rightarrow$ $\widetilde{\Theta}_2'$ - состоятельная

c) Сравнение эффективности оценок.

$\widetilde{\Theta}_1 = \frac{2}{3}\overline{X}$; $\widetilde{\Theta}_2' = \frac{n+1}{2n+1} x_{max}$; $D\widetilde{\Theta}_1 = \frac{\Theta^2}{27n}$; $D\widetilde{\Theta}_2' = \frac{n\Theta^2}{(2n+1)^2(n+2)}$

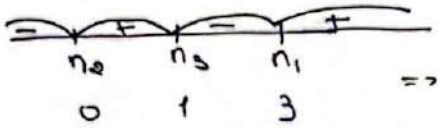
$\frac{1}{27n} \overset{?}{} \frac{n}{(2n+1)^2(n+2)}$

$27n^2 \quad (4n^2+4n+1)(n+2)$

$4n^3 - 15n^2 + 3n + 2 \overset{?}{} 0$

$(n-1)(4n^2-11n-2)$; $4n^2-11n-2=0$; $n_{12} = \frac{11}{8} \pm \frac{3}{8}\sqrt{17}$

$n_1 \approx 3$
$n_2 \approx 0$ - округляем до целого значения



$\Rightarrow$ после $n=3$: $D\widetilde{\Theta}_1 \geq D\widetilde{\Theta}_2'$

$\Rightarrow$ $\widetilde{\Theta}_2'$ - эффективнее $\widetilde{\Theta}_1$

d) Точный доверительный интервал для $\widetilde{\Theta}_2$ (β)

$f(\vec{x}_n, h) = \frac{x_{max}}{\Theta} - 1$ $\qquad F(x,\theta) = (\frac{x}{\theta} - 1)\mathbb{1}(\theta; 2\theta)\}$

$P(f<t) = P(x_{max} < \Theta t + \Theta) = (F(\Theta t + \Theta))^n = (t+1-1)^n \mathbb{1}(\theta; 2\theta)\} =$

$= t^n \mathbb{1}(\theta; 2\theta)\}$

t_1: $F(t_1) = \frac{1-\beta}{2}$; t_2: $F(t_2) = \frac{1+\beta}{2}$ $\Big\}$: $\sqrt[n]{\frac{1-\beta}{2}} < \frac{x_{max}}{\Theta} - 1 < \sqrt[n]{\frac{1+\beta}{2}}$

$P\left(\sqrt[n]{\frac{1-\beta}{2}} < \frac{x_{max}}{\Theta} - 1 < \sqrt[n]{\frac{1+\beta}{2}}\right) = \beta$

$\sqrt[n]{\frac{1-\beta}{2}} + 1 < \frac{x_{max}}{\Theta} < \sqrt[n]{\frac{1+\beta}{2}} + 1 \Rightarrow \frac{x_{max}}{\sqrt[n]{\frac{1+\beta}{2}} + 1} < \Theta < \frac{x_{max}}{\sqrt[n]{\frac{1-\beta}{2}} + 1}$

e) Асимптотический доверит. интервал.

ОММ: $\widetilde{\Theta}_1 = \frac{2}{3}\widetilde{\nu}_1 = g(\widetilde{\alpha}_1)$; $\nabla g(\alpha) = \frac{2}{3}$; $\Theta = g(\alpha)$; $K_{11} = \alpha_{1+1} - \alpha_1\alpha_1$

$\sqrt{n} \cdot \frac{g(\widetilde{\nu}) - g(\alpha)}{\sigma(\widetilde{\alpha})} \sim N(0;1)$ $\qquad \sigma^2(\alpha) = \nabla^T g(\alpha) K \nabla g(\alpha)$

$\sigma^2(\alpha) = \frac{2}{3}(\alpha_2 - \alpha_1^2)\frac{2}{3} = \frac{4}{9}(\alpha_2 - \alpha_1^2)$

$\frac{\sqrt{n}(\widetilde{\Theta}_1 - \Theta)}{\frac{2}{3}\sqrt{\widetilde{\alpha}_2 - \widetilde{\alpha}_1^2}} \sim N(0;1)$; $u_{\frac{1-\beta}{2}} = t_1$, $u_{\frac{1+\beta}{2}} = t_2$ $\Rightarrow P\left(t_1 < \frac{\sqrt{n}(\widetilde{\Theta}_1 - \Theta)}{\frac{2}{3}\sqrt{\widetilde{\alpha}_2 - \widetilde{\alpha}_1^2}} < t_2\right) = \beta$

$t_1 \cdot \frac{2}{3} \cdot \frac{\sqrt{\widetilde{\alpha}_2 - \widetilde{\alpha}_1^2}}{\sqrt{n}} < \widetilde{\Theta}_1 - \Theta < t_2 \cdot \frac{2}{3} \frac{\sqrt{\widetilde{\alpha}_2 - \widetilde{\alpha}_1^2}}{\sqrt{n}} \Rightarrow \widetilde{\Theta}_1 - \frac{2}{3}\frac{\sqrt{\widetilde{\alpha}_2 - \widetilde{\alpha}_1^2}}{\sqrt{n}} t_2 < \Theta < \widetilde{\Theta}_1 - \frac{2}{3} \cdot \frac{\sqrt{\widetilde{\alpha}_2 - \widetilde{\alpha}_1^2}}{\sqrt{n}} t_1$